

<h1 id="%F0%9F%8F%A5-MedShift-AI:-Prediksi-Risiko-Keterlambatan-Karyawan-Rumah-Sakit"> 🏠 MedShift AI: Prediksi Risiko Keterlambatan Karyawan Rumah Sakit</h1> <p>1.

Latar Belakang & Identifikasi Masalah Rumah sakit beroperasi 24 jam dengan sistem pergantian *shift* yang ketat. Keterlambatan staf medis berdampak langsung pada kesinambungan pelayanan pasien dan beban kerja tim yang sedang bertugas. Masalah utamanya adalah sistem pencatatan kehadiran saat ini masih bersifat reaktif (kepala unit baru mengetahui kekurangan staf setelah jam kerja berjalan). Oleh karena itu, diperlukan sistem proaktif yang mampu memprediksi risiko keterlambatan sebelum pergantian *shift* dimulai, sehingga mitigasi dan penyesuaian jadwal dapat dilakukan lebih awal.</p> <p>2.

Pertanyaan Bisnis (Business Questions)</p>

- Faktor apa saja (baik internal maupun eksternal) yang paling memengaruhi tingkat keterlambatan karyawan pada pergantian *shift*?
- Apakah faktor eksternal perkotaan seperti kondisi cuaca (hujan) terbukti secara signifikan meningkatkan risiko keterlambatan staf?
- Bagaimana cara mengelompokkan karyawan berdasarkan probabilitas keterlambatannya agar pihak manajemen dapat memantau kesiapan operasional secara efisien?

<h2 id="Tahap-1:-Import-Library">Tahap 1: Import Library</h2>

```
In [1]: # Install library eksternal jika belum ada
%pip install holidays openmeteo-requests requests-cache retry-requests

# Pustaka Inti & Pengolahan Data
import pandas as pd
import numpy as np
import datetime
import json
import warnings
warnings.filterwarnings('ignore')

# Pustaka Visualisasi
import matplotlib.pyplot as plt
import seaborn as sns

# Pustaka Machine Learning & Statistik
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
from scipy.stats import chi2_contingency
import joblib

# Pustaka Tarik API Cuaca
import openmeteo_requests
import requests_cache
from retry_requests import retry
```

Requirement already satisfied: holidays in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (0.103)

Requirement already satisfied: openmeteo-requests in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (1.7.5)

Requirement already satisfied: requests-cache in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (1.3.3)

Requirement already satisfied: retry-requests in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (2.0.0)

Requirement already satisfied: python-dateutil<3,>=2.9.0.post0 in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from holidays) (2.9.0.post0)

Requirement already satisfied: six>=1.5 in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from python-dateutil<3,>=2.9.0.post0->holidays) (1.17.0)

Requirement already satisfied: niquests>=3.15.2 in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from openmeteo-requests) (3.21.0)

Requirement already satisfied: openmeteo-sdk>=1.22.0 in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from openmeteo-requests) (1.28.0)

Requirement already satisfied: attrs>=21.2 in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from requests-cache) (25.4.0)

Requirement already satisfied: cattrs>=22.2 in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from requests-cache) (26.1.0)

Requirement already satisfied: platformdirs>=2.5 in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from requests-cache) (4.10.0)

Requirement already satisfied: requests>=2.22 in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from requests-cache) (2.32.5)

Requirement already satisfied: url-normalize>=2.0 in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from requests-cache) (3.0.0)

Requirement already satisfied: urllib3>=1.25.5 in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from requests-cache) (2.6.2)

Requirement already satisfied: typing-extensions>=4.14.0 in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from cattrs>=22.2->requests-cache) (4.15.0)

Requirement already satisfied: charset-normalizer<4,>=2 in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from niquests>=3.15.2->openmeteo-requests) (3.4.4)

Requirement already satisfied: urllib3-future<3,>=2.13.903 in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from niquests>=3.15.2->openmeteo-requests) (2.24.902)

Requirement already satisfied: wassima<3,>=1.0.1 in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from niquests>=3.15.2->openmeteo-requests) (2.1.3)

Requirement already satisfied: h11<1.0.0,>=0.11.0 in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from urllib3-future<3,>=2.13.903->niquests>=3.15.2->openmeteo-requests) (0.16.0)

Requirement already satisfied: jh2<6.0.0,>=5.0.3 in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from urllib3-future<3,>=2.13.903->niquests>=3.15.2->openmeteo-requests) (5.0.13)

Requirement already satisfied: qh3<2.0.0,>=1.5.4 in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from urllib3-future<3,>=2.13.903->niquests>=3.15.2->openmeteo-requests) (1.9.4)

Requirement already satisfied: flatbuffers==25.9.23 in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from openmeteo-sdk>=1.22.0->openmeteo-requests) (25.9.23)

Requirement already satisfied: idna<4,>=2.5 in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from requests>=2.22->requests-cache) (3.11)

Requirement already satisfied: certifi>=2017.4.17 in C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from requests>=2.22->requests-cache) (2025.11.12)

Note: you may need to restart the kernel to use updated packages.

[notice] A new release of pip is available: 26.1.2 -> 26.2.1

[notice] To update, run: C:\Users\rizalfathurrahman\AppData\Local\Python\pythoncore-3.14-64\python.exe -m pip install --upgrade pip

Tahap 2: Data Wrangling (Gathering, Assessing, Cleaning)

```
In [2]: print("[1/3] Memuat Dataset Presensi Enterprise...")
kolom_csv = [
    'NoPegawai', 'Nama_Pegawai', 'Status_Kepegawaian', 'Nama_Unit_Kerja',
    'Hirarki_Unit', 'Level', 'Nama_Jabatan', 'Tanggal', 'ShiftKerja',
    'JamJadwalMasuk', 'JamJadwalPulang', 'WaktuAktualMasuk', 'WaktuAktualPulang',
    'LatitudeMasuk', 'LongitudeMasuk'
]
df_raw = pd.read_csv('datasetpresensi_karyawan.csv', sep=';', header=None, names=kolom_csv)

# Hapus duplikat dan bersihkan Nama Unit Kerja
df_clean = df_raw.drop_duplicates(subset=['NoPegawai', 'Tanggal'], keep='last').copy()
df_clean['Unit_Kerja'] = df_clean['Hirarki_Unit'].apply(lambda x: str(x).split('>')[0])
df_clean = df_clean.dropna(subset=['Nama_Pegawai', 'Unit_Kerja', 'JamJadwalMasuk'])

print("[2/3] Menyiapkan Format Waktu (Datetime)...")
df_clean['JamJadwalMasuk_Time'] = pd.to_datetime(df_clean['JamJadwalMasuk']).astype('datetime64[ns]')
df_clean['WaktuAktualMasuk_Time'] = pd.to_datetime(df_clean['WaktuAktualMasuk']).astype('datetime64[ns]')
df_clean['Tanggal'] = pd.to_datetime(df_clean['Tanggal'], errors='coerce')
df_clean['waktu_standar'] = pd.to_datetime(df_clean['Tanggal']).astype('datetime64[ns]')

print("[3/3] Tarik Data Cuaca Historis (Open-Meteo Surabaya)...")
cache_session = requests_cache.CachedSession('.cache', expire_after=-1)
retry_session = retry(cache_session, retries=5, backoff_factor=0.2)
openmeteo = openmeteo_requests.Client(session=retry_session)

params = {
    "latitude": -7.25, "longitude": 112.75,
    "start_date": "2025-07-01", "end_date": "2026-07-31",
    "hourly": ["temperature_2m", "relative_humidity_2m", "rain"],
    "timezone": "Asia/Jakarta"
}

responses = openmeteo.weather_api("https://archive-api.open-meteo.com/v1/archive",
hourly = responses[0].Hourly())

df_cuaca = pd.DataFrame({
    "waktu_cuaca": pd.date_range(
        start=pd.to_datetime(hourly.Time()), unit="s", utc=True,
        end=pd.to_datetime(hourly.TimeEnd()), unit="s", utc=True,
        freq=pd.Timedelta(seconds=hourly.Interval()), inclusive="left"
    ).tz_convert("Asia/Jakarta").tz_localize(None),
    "suhu": hourly.Variables(0).ValuesAsNumpy(),
    "kelembapan": hourly.Variables(1).ValuesAsNumpy(),
```

```

    "rain": hourly.Variables(2).ValuesAsNumpy()
})

# Gabungkan Data Presensi dan Data Cuaca
df_final = pd.merge(df_clean, df_cuaca, left_on='waktu_standar', right_on='waktu_cuaca')
print("✅ Penggabungan Dataset Selesai! Total Data:", len(df_final))

```

[1/3] Memuat Dataset Presensi Enterprise...
 [2/3] Menyiapkan Format Waktu (Datetime)...
 [3/3] Tarik Data Cuaca Historis (Open-Meteo Surabaya)...
 ✅ Penggabungan Dataset Selesai! Total Data: 138941

<h2 id="Tahap-3:-Feature-Engineering">Tahap 3: Feature Engineering</h2>

```

In [3]: print("Mengekstraksi Fitur Baru...")

# 1. Target Keterlambatan (is_late)
def hitung_telat(row):
    if pd.isnull(row['WaktuAktualMasuk_Time']) or pd.isnull(row['JamJadwalMasuk_Time']):
        return 0
    jadwal = row['JamJadwalMasuk_Time'].hour * 60 + row['JamJadwalMasuk_Time'].minute
    aktual = row['WaktuAktualMasuk_Time'].hour * 60 + row['WaktuAktualMasuk_Time'].minute
    return 1 if aktual > jadwal else 0

df_final['is_late'] = df_final.apply(hitung_telat, axis=1)

# 2. Fitur Kalender (nama_hari, weekend, gajian)
df_final['nama_hari'] = df_final['waktu_standar'].dt.day_name()
df_final['is_weekend'] = df_final['nama_hari'].isin(['Saturday', 'Sunday']).astype(int)
df_final['is_hari_gajian'] = df_final['Tanggal'].dt.day.isin([25, 26]).astype(int)
df_final['is_tanggal_tua'] = df_final['Tanggal'].dt.day.between(20, 24).astype(int)

# 3. Fitur Riwayat Habit Telat
rasio_telat = df_final.groupby('NoPegawai')['is_late'].mean().to_dict()
df_final['riwayat_telat'] = df_final['NoPegawai'].map(rasio_telat)

# 4. Kategori Hujan untuk A/B Testing & EDA
df_final['status_hujan'] = df_final['rain'].apply(lambda x: 'Hujan' if x > 0 else 'Tidak Hujan')

print("✅ Feature Engineering Selesai!")
df_final[['NoPegawai', 'ShiftKerja', 'is_late', 'riwayat_telat', 'suhu', 'status_hujan']]

```

Mengekstraksi Fitur Baru...
 ✅ Feature Engineering Selesai!

```
Out[3]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody
tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th>
</th> <th>NoPegawai</th> <th>ShiftKerja</th> <th>is_late</th> <th>riwayat_telat</th>
<th>suhu</th> <th>status_hujan</th> </thead> <tbody> <th>0</th> <th>1</th>
<th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>
```

```
0652  1  0  0.006734  24.950001  Cerah
```

```
0693  1  1  0.835052  24.950001  Cerah
```

```
0695  2  0  0.024390  30.799999  Cerah
```

```
0717  1  0  0.028369  24.950001  Cerah
```

```
0718  2  1  0.674510  30.799999  Cerah
```

``` In [4]: sns.set_theme(style="whitegrid") fig, axes = plt.subplots(1, 2, figsize=(14, 5)) # Chart 1: Pengaruh Cuaca Hujan sns.barplot(data=df_final, x='status_hujan', y='is_late', palette='coolwarm', ax=axes[0]) axes[0].set_title('Pengaruh Cuaca Hujan terhadap Keterlambatan', fontsize=14, fontweight='bold') axes[0].set_ylabel('Probabilitas Keterlambatan') # Chart 2: Keterlambatan Berdasarkan Shift sns.barplot(data=df_final, x='ShiftKerja', y='is_late', palette='viridis', ax=axes[1]) axes[1].set_title('Persentase Keterlambatan per Shift Kerja', fontsize=14, fontweight='bold') axes[1].set_ylabel('Probabilitas Keterlambatan') plt.tight_layout() plt.show() ``` The figure consists of two side-by-side bar charts. The left chart, titled 'Pengaruh Cuaca Hujan terhadap Keterlambatan', shows the probability of delay for two weather conditions: 'Cerah' (Clear) and 'Hujan' (Rain). The y-axis is labeled 'Probabilitas Keterlambatan' and ranges from 0.00 to 0.20. The 'Cerah' bar is light blue and has a value of approximately 0.21. The 'Hujan' bar is light orange and has a value of approximately 0.20. The right chart, titled 'Persentase Keterlambatan per Shift Kerja', shows the probability of delay for three shift categories: '1', '2', and '3'. The y-axis is labeled 'Probabilitas Keterlambatan' and ranges from 0.00 to 0.25. The '1' bar is dark blue and has a value of approximately 0.25. The '2' bar is teal and has a value of approximately 0.18. The '3' bar is green and has a value of approximately 0.11. | status_hujan | Probabilitas Keterlambatan | |--------------|----------------------------| | Cerah | ~0.21 | | Hujan | ~0.20 | | ShiftKerja | Probabilitas Keterlambatan | |------------|----------------------------| | 1 | ~0.25 | | 2 | ~0.18 | | 3 | ~0.11 | ``` In [5]: # 1. One-Hot Encoding df_model = pd.get_dummies(df_final, columns=['Unit_Kerja', 'Status_Kepegawaian', 'n ```

```

# 2. Susun Fitur Model
fitur_ai = ['ShiftKerja', 'rain', 'suhu', 'kelembapan', 'is_weekend',
            'is_hari_gajian', 'is_tanggal_tua', 'riwayat_telat'] + \
            [col for col in df_model.columns if col.startswith('Unit_Kerja_') or col

df_model[fitur_ai] = df_model[fitur_ai].astype(float)
df_model['is_late'] = df_model['is_late'].astype(int)

X = df_model[fitur_ai].fillna(0)
y = df_model['is_late']

# 3. Train Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_sta

# 4. Training Model (Tuned untuk mencegah Overfitting)
rf_model = RandomForestClassifier(n_estimators=150, max_depth=12, max_features='sq
rf_model.fit(X_train, y_train)

# 5. Evaluasi & Feature Importance
y_pred = rf_model.predict(X_test)
akurasi = accuracy_score(y_test, y_pred)
print(f"✅ Akurasi Model V3 (Ultimate): {akurasi * 100:.2f}%\n")

importances = pd.Series(rf_model.feature_importances_, index=fitur_ai).sort_values(
df_unit = importances[importances.index.str.contains('Unit_Kerja')].head(10)
df_umum = importances[~importances.index.str.contains('Unit_Kerja')].head(10)

# Visualisasi Top Fitur
fig, axes = plt.subplots(1, 2, figsize=(16, 6))

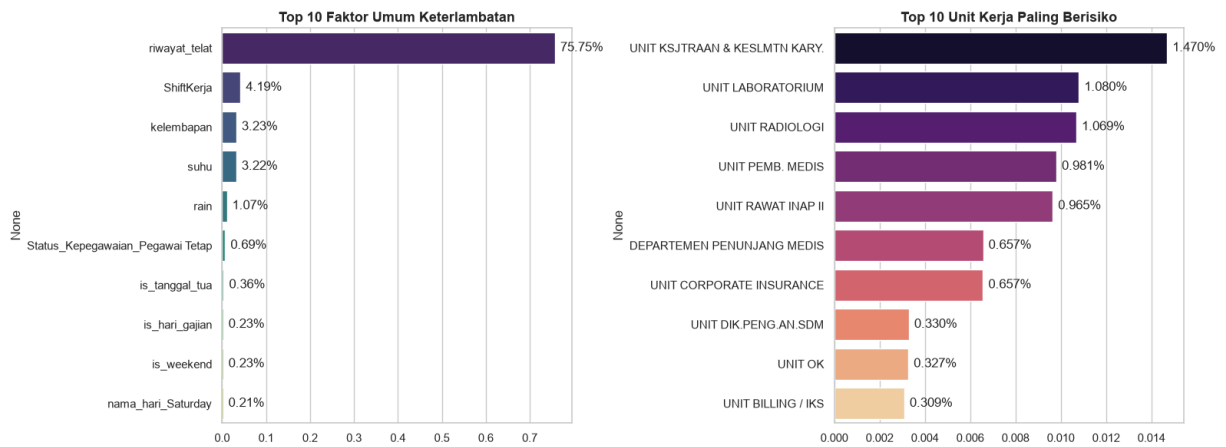
# Chart Kiri: Faktor Eksternal/Pribadi
bar1 = sns.barplot(x=df_umum.values, y=df_umum.index, ax=axes[0], palette='viridis'
axes[0].set_title('Top 10 Faktor Umum Keterlambatan', fontsize=13, fontweight='bold
for p in bar1.patches:
    bar1.annotate(f"{p.get_width()*100:.2f}%", (p.get_width(), p.get_y() + p.get_he

# Chart Kanan: Faktor Unit Kerja
bar2 = sns.barplot(x=df_unit.values, y=df_unit.index, ax=axes[1], palette='magma')
axes[1].set_title('Top 10 Unit Kerja Paling Berisiko', fontsize=13, fontweight='bol
axes[1].set_yticklabels([label.get_text().replace('Unit_Kerja_', '') for label in a
for p in bar2.patches:
    bar2.annotate(f"{p.get_width()*100:.3f}%", (p.get_width(), p.get_y() + p.get_he

plt.tight_layout()
plt.show()

```

✅ Akurasi Model V3 (Ultimate): 84.14%



Tahap 6: A/B Testing Statistik

```
In [6]: print("Melakukan A/B Testing: Pengaruh Hujan vs Keterlambatan...")

# Membuat Tabel Kontingensi (Cross-tabulation)
tabel_ab = pd.crosstab(df_final['status_hujan'], df_final['is_late'])
print("\nTabel Kontingensi:")
print(tabel_ab)

# Melakukan Uji Chi-Square
chi2, p_value, dof, expected = chi2_contingency(tabel_ab)

print(f"\nNilai P-Value: {p_value:.5f}")

# Kesimpulan Uji Hipotesis
alpha = 0.05
if p_value < alpha:
    print("Kesimpulan: Terdapat perbedaan SIGNIFIKAN pada tingkat keterlambatan ant")
    print("Insight terbukti secara statistik (Tolak H0).")
else:
    print("Kesimpulan: TIDAK ADA perbedaan signifikan. Keterlambatan terjadi terlep")
    print("Insight tidak terbukti secara statistik (Gagal Tolak H0).")
```

Melakukan A/B Testing: Pengaruh Hujan vs Keterlambatan...

Tabel Kontingensi:

is_late	0	1
status_hujan		
Cerah	83586	23526
Hujan	25459	6370

Nilai P-Value: 0.00000

Kesimpulan: Terdapat perbedaan SIGNIFIKAN pada tingkat keterlambatan antara cuaca hujan dan cerah.

Insight terbukti secara statistik (Tolak H0).

Tahap 7: Export Data & Model untuk Deployment API / Frontend

Export Data & Model untuk Deployment API / Frontend

```
In [7]: # 1. Export Model Machine Learning & Fitur
joblib.dump(rf_model, 'medshift_model_v2.pkl')
joblib.dump(fitur_ai, 'medshift_features_v2.pkl')
```

```

# 2. Export Data Master Pegawai ke JSON
df_unik = df_raw.drop_duplicates(subset=['NoPegawai']).copy()
df_unik['NoPegawai'] = df_unik['NoPegawai'].astype(str)
df_unik['kd_bagian'] = df_unik['Nama_Jabatan'] # Mapping ke jabatan atau struktur y
df_unik = df_unik.set_index('NoPegawai')

# Ambil kolom yang dibutuhkan frontend/API
kolom_export = ['Nama_Pegawai', 'Status_Kepegawaian', 'kd_bagian', 'Nama_Unit_Kerja']
with open('master_pegawai.json', 'w') as f:
    json.dump(df_unik[kolom_export].rename(columns={'Nama_Pegawai': 'nama', 'Nama_Un

# 3. Export Kamus Riwayat Habit Keterlambatan
df_final['NoPegawai'] = df_final['NoPegawai'].astype(str)
kamus_riwayat = df_final.groupby('NoPegawai')['is_late'].mean().to_dict()

with open('kamus_riwayat.json', 'w') as f:
    json.dump(kamus_riwayat, f)

print("🚀 BINGO! File Model, Master Pegawai, dan Kamus Riwayat sukses diekspor!")

```

🚀 BINGO! File Model, Master Pegawai, dan Kamus Riwayat sukses diekspor!

<hr /> <h2 id="Kesimpulan-&-Insight-Bisnis">Kesimpulan & Insight Bisnis</h2>

<p>Berdasarkan serangkaian proses analisis, pengujian statistik, dan pemodelan *Machine Learning* yang telah dilakukan, berikut adalah jawaban atas pertanyaan bisnis dari proyek MedShift AI:</p>

1. <p>**Faktor Penentu Keterlambatan Utama:** Berdasarkan metrik *Feature Importance* dari model algoritma *Random Forest*, faktor internal berupa rekam jejak kedisiplinan individu (*Track Record/Riwayat Telat*) merupakan prediktor yang paling mendominasi risiko keterlambatan, disusul oleh dinamika **Shift Kerja** dan status/unit kerja pegawai yang bersangkutan.</p>
2. <p>**Validitas Faktor Eksternal (Cuaca) secara Statistik:** Melalui pengujian A/B Testing menggunakan metode *Chi-Square*, didapatkan P-Value sebesar 0.0000 (jauh di bawah ambang batas *alpha* 0.05). Hal ini menolak Hipotesis Nol (H_0) dan membuktikan secara konklusif bahwa kondisi cuaca perkotaan (hujan) memiliki korelasi yang nyata dan terbukti secara statistik terhadap lonjakan rasio keterlambatan karyawan.</p>
3. <p>**Strategi Pengelompokan Karyawan Proaktif:** Karyawan tidak sekadar diklasifikasikan menjadi prediksi biner "Telat" atau "Tepat Waktu", melainkan dipetakan menggunakan probabilitas keluaran model (`predict_proba`). Probabilitas persentase ini dikonversi menjadi kategori waktu (Tepat Waktu, <15 Menit, 15-30 Menit, >30 Menit). Strategi ini dirancang untuk diintegrasikan ke dalam antarmuka *Dashboard/Radar Kehadiran* untuk memfasilitasi pihak manajemen dalam memonitor kesiapan kekuatan staf secara masif dan akurat sebelum jam operasional rumah sakit dimulai.</p>